



# JERUSALEM COLLEGE OF ENGINEERING

## AN AUTONOMOUS INSTITUTION

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### DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

|                          |   |                             |
|--------------------------|---|-----------------------------|
| Regulation               | : | 2023                        |
| Course Name              | : | DATABASE MANAGEMENT SYSTEMS |
| Course Code              | : | JCS2401                     |
| Year / Semester / Branch | : | II / IV / CSE               |
| Unit No                  | : | 3                           |
| Unit Title               | : | TRANSACTION PROCESSING      |
| No. of Hours             | : | 9                           |
| Weightage in Exam (%)    | : | 20%                         |

### QUESTION BANK

#### PART A

| Q.No. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks | BTL | CO  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| 1.    | <b>Define transaction.</b> <ul style="list-style-type: none"><li>• A transaction is a set of operations executed as a single logical unit that accesses and may modify database data.</li><li>• It follows the all-or-nothing rule and ensures data integrity and consistency during execution.</li></ul>                                                                                                                                                                                                                                                                                  | 2     | K1  | CO3 |
| 2.    | <b>List out the various states of a transaction. Give the diagrammatic representation of the same.</b> <p><b>Active</b> – the initial state; the transaction stays in this state while it is executing</p> <p><b>Partially committed</b> – after the final statement has been executed.</p> <p><b>Failed</b> - after the discovery that normal execution can no longer proceed.</p> <p><b>Aborted</b> – after the transaction has been rolled back and the database restored to its state prior to the start of the transaction.</p> <p><b>Committed</b> – after successful completion</p> | 2     | K1  | CO3 |



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|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|    | <pre> graph LR     Active((Active)) --&gt; PartiallyCommitted((Partially Committed))     Active --&gt; Failed((Failed))     PartiallyCommitted --&gt; Committed((Committed))     Failed --&gt; Aborted((Aborted))           </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| 3. | <p><b>How do schedules ensure consistency when managing concurrent transactions in DBMS?</b></p> <p>Schedule – a sequence of instructions that specifies the chronological order in which instructions of concurrent transactions are executed.</p> <p>A schedule for a set of transactions must consist of all instructions of those transactions must preserve the order in which the instructions appear in each transaction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2 | K2 | CO3 |
| 4. | <p><b>Identify the function of serializability? Give its various forms.</b></p> <p>Serializability ensures that the execution of multiple transactions gives the same result as if they were executed one after another (serially). Its main function is to maintain data consistency in the database. It helps in avoiding conflicts when transactions run at the same time. It also ensures that the database remains correct and reliable after concurrent execution. Different forms of schedule equivalence give rise to the notions of:</p> <ul style="list-style-type: none"> <li>• Conflict serializability</li> <li>• View serializability</li> </ul>                                                                                                                                                                                                                | 2 | K2 | CO3 |
| 5. | <p><b>Illustrate when two instructions are said to be in conflict in transaction?</b></p> <p>Instructions <math>li</math> and <math>lj</math> of transactions <math>Ti</math> and <math>Tj</math>, respectively, conflict if and only if there exists some item <math>Q</math> accessed by both <math>li</math> and <math>lj</math>, and at least one of these instructions wrote <math>Q</math>.</p> <ul style="list-style-type: none"> <li>• <math>li = \text{read}(Q)</math>, <math>lj = \text{read}(Q)</math>. <math>li</math> and <math>lj</math> don't conflict.</li> <li>• <math>li = \text{read}(Q)</math>, <math>lj = \text{write}(Q)</math>. They conflict.</li> <li>• <math>li = \text{write}(Q)</math>, <math>lj = \text{read}(Q)</math>. They conflict</li> <li>• <math>li = \text{write}(Q)</math>, <math>lj = \text{write}(Q)</math>. They conflict</li> </ul> | 2 | K2 | CO3 |
| 6. | <p><b>State the property of conflict serializability.</b></p> <p>If a schedule <math>S</math> can be transformed into a schedule <math>S'</math> by a series of swaps of non-conflicting instructions, we say that <math>S</math> and <math>S'</math> are</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 | K1 | CO3 |



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|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|     | conflict equivalent and that schedule $S$ is conflict serializable if it is conflict equivalent to a serial schedule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |     |
| 7.  | <p><b>Outline the conditions for view serializability.</b></p> <ul style="list-style-type: none"> <li>Let <math>S</math> and <math>S'</math> be two schedules with the same set of transactions. <math>S</math> and <math>S'</math> are view equivalent if the following three conditions are met, for each data item <math>Q</math>,</li> <li>If in schedule <math>S</math>, transaction <math>T_i</math> reads the initial value of <math>Q</math>, then in schedule <math>S'</math> also transaction <math>T_i</math> must read the initial value of <math>Q</math>.</li> <li>If in schedule <math>S</math> transaction <math>T_i</math> executes <math>\text{read}(Q)</math>, and that value was produced by transaction <math>T_j</math> (if any), then in schedule <math>S'</math> also transaction <math>T_i</math> must read the value of <math>Q</math> that was produced by the same <math>\text{write}(Q)</math> operation of transaction <math>T_j</math>.</li> <li>The transaction (if any) that performs the final <math>\text{write}(Q)</math> operation in schedule <math>S</math> must also perform the final <math>\text{write}(Q)</math> operation in schedule <math>S'</math>.</li> </ul> | 2 | K2 | CO3 |
| 8.  | <p><b>Identify the different modes of lock in a transaction.</b></p> <p>Lock is the most commonly used to implement the requirement is to allow a transaction to access a data item only if it is currently holding a lock on that item.</p> <p>Data items can be locked in two modes:</p> <ul style="list-style-type: none"> <li>Exclusive (X) mode. Data items can be both read as well as written. X-lock is requested using lock-X instruction.</li> <li>Shared (S) mode. Data item can only be read. S-lock is requested using lock-S instruction.</li> </ul> <p>Lock requests are made to concurrency-control manager. Transaction can proceed only after request is granted.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2 | K1 | CO3 |
| 9.  | <p><b>How does a transaction locking protocol work ?</b></p> <ul style="list-style-type: none"> <li>A transaction locking protocol controls access to data by assigning <b>locks</b> (shared or exclusive) before operations are performed.</li> <li>It ensures <b>concurrent transactions do not interfere</b>, maintaining data consistency by allowing only safe access and releasing locks after execution..</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2 | K2 | CO3 |
| 10. | <b>Why are ACID properties crucial for transaction management in databases?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2 | K2 | CO3 |



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|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|     | <ul style="list-style-type: none"> <li>● <b>Atomicity</b> - Atomicity means that an entire transaction either takes place all at once or it doesn't occur at all. It means that there's no midway</li> <li>● <b>Consistency</b> - Consistency means that we have to maintain the integrity constraints so that any given database stays consistent both before and after a transaction.</li> <li>● <b>Isolation</b> - Isolation ensures the occurrence of multiple transactions concurrently without a database state leading to a state of inconsistency</li> <li>● <b>Durability</b> - The durability property states that once the execution of a transaction is completed, the modifications and updates on the database gets written on and stored in the disk.</li> </ul> |   |    |     |
| 11. | <p><b>Classify the phases of two-phase locking protocol.</b></p> <p><b>Growing phase:</b> a transaction may obtain locks but not release any lock.</p> <p><b>Shrinking phase:</b> a transaction may release locks but may not obtain any new locks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 | K2 | CO3 |
| 12. | <p><b>Define Deadlock with an example.</b></p> <p>A deadlock occurs when two or more tasks permanently block each other by each task having a lock on a resource that the other tasks are trying to lock.</p> <p>For example, let us assume we have two processes, P1 and P2. Now, process P1 is holding the resource R1 and is waiting for the resource R2. At the same time, process P2 is having the resource R2 and is waiting for the resource R1.</p>                                                                                                                                                                                                                                                                                                                     | 2 | K1 | CO3 |
| 13. | <p><b>Differentiate between the strict two-phase locking protocol and the rigorous two-phase locking protocol.</b></p> <ul style="list-style-type: none"> <li>● <b>Strict two-phase locking:</b> <ul style="list-style-type: none"> <li>□ It holds all it is exclusive that is write locks until commit time, one time granted.</li> <li>□ It locks all its items before it starts thus once transaction starts, it is in its shrinking phase</li> </ul> </li> <li>● <b>Rigorous two-phase locking:</b></li> </ul>                                                                                                                                                                                                                                                              | 2 | K2 | CO3 |



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|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|     | <ul style="list-style-type: none"> <li>□ It is more restrictive as compared to standard 2PL in that it enforces the rule that no locks can be released through a transaction until after it commits or aborts.</li> <li>□ It does not unlock any of its items until after it terminates to the transaction is in its expanding phase till it ends.</li> </ul>                                                                                                                                                                                                                                                                            |   |    |     |
| 14. | <p><b>Interpret the working of Save point and Rollback commands in SQL.</b></p> <p>A Save point is a logical rollback point within a transaction. When you set a save point, whenever an error occurs past a save point, you can undo the events you have done up to the save point using the rollback.</p> <p><b>Syntax:</b><br/>begin;<br/>savepoint &lt;savepoint name&gt;;</p> <p>Rollback command serves only in the creation of a SAVEPOINT among all the transactional statements. The ROLLBACK command is used to undo a group of transactions.</p> <p><b>Syntax :</b> Rollback to &lt;savepoint _name&gt;;</p>                  | 2 | K2 | CO3 |
| 15. | <p><b>Specify different isolation levels in the database</b></p> <ul style="list-style-type: none"> <li>● Serializable.</li> <li>● Repeatable reads.</li> <li>● Read committed.</li> <li>● Read uncommitted.</li> <li>● Dirty reads.</li> <li>● Non-repeatable reads.</li> </ul>                                                                                                                                                                                                                                                                                                                                                         | 2 | K2 | CO3 |
| 16. | <p><b>List and define the two approaches for deadlock prevention.</b></p> <ul style="list-style-type: none"> <li>● <b>Wait–Die Scheme:</b><br/>A non-pre-emptive technique based on timestamps where an older transaction is allowed to wait for a younger one, but a younger transaction requesting a resource held by an older one is aborted (dies) and restarted later.</li> <li>● <b>Wound–Wait Scheme:</b><br/>A pre-emptive technique where an older transaction pre-empt (wounds) a younger one, forcing the younger transaction to abort, while a younger transaction waits if the resource is held by an older one.</li> </ul> | 2 | K1 | CO3 |



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|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 17. | <b>Why is a wait-for <i>graph</i> important in Deadlock?</b><br>The wait-for graph is crucial for detecting and resolving deadlocks efficiently, ensuring smooth transaction processing and maintaining database consistency.                                                                                                                                                                                    | 2 | K2 | CO3 |
| 18. | <b>Name the two operations involved in recovery</b> <ul style="list-style-type: none"> <li>• <b>Undo (TRX):</b> This command is used to restore all records updated by transactions to the old value.</li> <li>• <b>Redo (TRX):</b> This command is used to set the value of all records updated by a transaction to the new value.</li> </ul>                                                                   | 2 | K1 | CO3 |
| 19. | <b>List the four conditions for deadlock to occur in a system.</b> <ul style="list-style-type: none"> <li>• mutual exclusion</li> <li>• hold and wait or partial allocation</li> <li>• no pre-emption</li> <li>• resource waiting or circular wait</li> </ul>                                                                                                                                                    | 2 | K1 | CO3 |
| 20. | <b>In a multi-user database system, how does isolation prevent issues like dirty reads, non-repeatable reads, and phantom reads?</b><br>Transaction isolation is an important part of any transactional system. It deals with consistency and completeness of data retrieved by queries un affecting a user data by other user actions. A database acquires locks on data to maintain a high level of isolation. | 2 | K2 | CO3 |

### PART B

| Q.No. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks | BTL | CO  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| 1.    | i) Consider the following schedules. The actions are listed in the order they are scheduled, and pre-fixed with the transaction name.<br><b>S1:</b> T1: R(X), T2: R(X), T1: W(Y), T2: W(Y), T1: R(X), T2: R(Y)<br><b>S2:</b> T3: W(X), T1: R(X), T1: W(Y), T2: R(Z), T2: W(Z), T3: R(Z)<br>For each of the schedules, analyse the following questions:<br>(a) Is the schedule conflict serializable? If so, what are all the conflict equivalent serial schedules?<br>(b) Is the schedule View serializable? If so, what are all the view equivalent serial schedules? | 8     | K4  | CO3 |



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|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----|-----|
|    | <p>ii) Consider the following two transactions:</p> <p><i>T1</i>: read(<i>A</i>);<br/>read(<i>B</i>);<br/><b>if</b> <i>A</i> = 0 <b>then</b> <i>B</i> := <i>B</i> + 1;<br/>write(<i>B</i>).<br/><i>T2</i>: read(<i>B</i>);<br/>read(<i>A</i>);<br/><b>if</b> <i>B</i> = 0 <b>then</b> <i>A</i> := <i>A</i> + 1;<br/>write(<i>A</i>).</p> <p>Add lock and unlock instructions to transactions <i>T1</i> and <i>T2</i>, so that they observe the two-phase locking protocol. Analyse if the execution of these transactions result in a deadlock?</p>                                                        | 8                 |    |     |
| 2. | Illustrate the process of Two-Phase Locking Protocol to a fund transfer transaction between two bank accounts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 16                | K2 | CO3 |
| 3. | <p>i) Two sellers update the same account balance of ₹10,000 concurrently:</p> <ul style="list-style-type: none"> <li>• <i>T1</i> withdraws ₹2,000</li> <li>• <i>T2</i> withdraws ₹3,000</li> </ul> <p>Evaluate how the lost update problem occurs in this scenario with example.</p> <p>ii) Transaction <i>T1</i> updates a product price from ₹500 to ₹450 but has not committed yet. Transaction <i>T2</i> reads this uncommitted value and displays it to customers. Later, <i>T1</i> is rolled back. Assess the consequences of dirty reads on database consistency and reliability with example.</p> | <p>8</p> <p>8</p> | K5 | CO3 |
| 4. | Analyse how the ARIES recovery algorithm handles a crash scenario.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 16                | K4 | CO3 |
| 5. | Explain how deadlocks can be managed in a high-concurrency database using both detection and prevention techniques.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 16                | K2 | CO3 |
| 6. | Evaluate the effectiveness of SQL facilities for managing Concurrency Control and Database Recovery in ensuring reliable and efficient transaction processing.                                                                                                                                                                                                                                                                                                                                                                                                                                             | 16                | K5 | CO3 |



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**COURSE FACULTY**

**COURSE COORDINATOR**

**HOD**

**SCRUTINIZER**